

Software Requirements Specification

Project: Compass — study-path advisor

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1. Introduction

<The introduction is an overview that allows people to understand the structure and usage principles of the software requirements specification. Outline briefly the content of the document, who it is directed to and other topics this document presents.>

1.1. Purpose

<This section contains the definition of the project discussed in this document. If this software requirements specification applies only to a part of the system, that part or subsystem should be identified. In other words, the problem that this project is intended to solve should be defined here. The problem definition can be reused.>

1.2. Intended audience

<The intended audience includes the description of readers for whom this document is intended. For example, it could be developers, project managers, users, testers, etc. The main goal of this section is to convey to the readers the information they need to know before reading this document.>

1.3. Additional information

<Anything else that needs to be mentioned and is not covered by the above subsections>

2. User requirements

<List all user requirements. These should describe the system as a complete product that, to certain extent, is a black-box which does certain tasks for the user. Link the user requirements with the high-level software objectives. Use statements written in natural language.>

3. System requirements

3.1. Functional requirements

What the system does for the student, independent of how it is built. Priority uses MoSCoW; “Traces to” links each function to the Task 1 sub-problem or objective it serves; “AI” marks the functions that depend on the AI component, which are carried forward into section 4.

ID	Feature / Function	Description	Priority	Action / Result	Traces to	AI
FR-1	Capture stated interest	Accept an interest or destination in the student’s own words, with no fixed vocabulary.	Must	Free text kept for the session.	1.2.4	No
FR-2	Capture study context	Record programme, year, completed courses, remaining credits and constraints (language, workload, exchange).	Must	Context stored for the session.	3.3	No
FR-3	Derive required skills	Determine the skills a stated destination demands, in a standard skills taxonomy.	Must	Ranked skills with confidence scores.	1.2.1	Yes
FR-4	Retrieve opportunities	Find courses, minors, projects, internships and exchanges that teach those skills.	Must	Candidates, each with the skills it adds.	1.2.2	Yes
FR-5	Assess feasibility	Exclude combinations that break prerequisites, credit rules, compulsory components or timetables.	Must	Infeasible options removed before ranking.	3.4	No
FR-6	Compose whole paths	Combine the remaining options into complete paths that fit the remaining study time.	Must	Whole paths, not isolated suggestions.	1.2.2	No
FR-7	Rank and return paths	Return three to five paths, ordered by fit to the stated interest.	Must	Ordered shortlist of 3–5 paths.	3.3	No
FR-8	State what a path opens	Name the destinations each path leads to: master’s programmes, research areas, roles.	Must	Outcome statement per path.	1.2.1	No
FR-9	State what a path closes off	Name what each path makes harder or rules out.	Should	Trade-off warning per path.	1.2.1	No
FR-10	Justify each path	Explain what each element contributes and why the paths are ordered as they are.	Should	Justification per path and element.	3.4	No
FR-11	Surface deadlines	Show which decisions must be taken by when, before an option expires.	Should	Timing signal per path.	1.2.3	No
FR-12	Re-evaluate on change	Recompute the remaining options when the student changes direction.	Should	Updated shortlist.	1.2.5	No
FR-13	Withhold low-confidence matches	Below the confidence threshold, return no match instead of a guess.	Must	Unmatched items shown as unmatched.	3.4	Yes

3.2. Non-functional requirements

<How are the functions defined in 3.1. going to be implemented. What are the constraints that would need to be met for the functions in 3.1. to be working. What external factors will impact the operation of the system. What ethical aspects would limit/expand the functionality of the system. And so on. Organise this section in a structured way and link it to the requirements in 3.1. May use a table or a list.>

4. AI specific requirements

<Indicate which requirements relate to the AI component and models and why.>